Software Project Management Plan

for

Library Automate

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1. Overview

1.1. Project Summary

1.1.1. Purpose, Scope and Objectives

The main purpose of Library Automate is to help the Trinidad Municipal College Library and Trinidad Public Library improved its services in terms of borrowing transaction and monitoring library users.

This project is developed to replace the old way of recording books in library, borrowers and books borrowed, and all library users. The old method of recording book borrowers is tedious since it is time consuming especially in writing the name of the borrower in the book and borrowers list, which also requires office resources. Another thing that adds problem to the staff is that they need to monitor the due date of every borrowing transaction.

The goal of Library Automate is to provide the following features required by the target users:

- Library Staff
  - Borrower's Information: System should be able to store borrower's information particularly from different sectors (Student, Faculty, and Outsiders).
  - Book Information: For storing book information that can be used in cataloging, and borrowing.
  - Transactions
    - Book Requisition: Registered borrowers has the privilege to access the system, wherein requesting book/s to be borrowed can be done in the system.

        - Book Issuance: For book borrowing transaction.
        - Return Book: For returning books borrowed.
    - Reports
        - Borrowing Transactions
        - Unreturned Books
        - Overdue Books
        - List of All Books
    - Library Users Monitoring
- Librarian
    - Dashboard: Summary of reports as part of librarian's monitoring (ex. library users, frequently used books, etc.)

However, the system is limited to the following:

- Fines for overdue books since the librarian doesn't require monetary value as sanction.
- The proposed project is intended only for Trinidad Municipal College Library and Trinidad Public Library.

1.1.2. Assumptions and Constraints

Library Automate is designed to be client-server setup. Therefore, the system will be built using the following technologies:

- Back-end: Hypertext Preprocessor
- Front-end: Bootstrap Framework
- Coding Pattern: MVC Architecture
- Database Server: MySQL

- Web Server: Apache

The project is set to finish on September 28, 2019 and subject for testing afterwards.

1.1.3. Project Deliverables

All of the items listed in this subsection are the deliverables that are to be provided prior to completion of the project.

- Software Program (Library Automate)
- Software Documentation (Software Engineering Docs)
  - Software Requirements Specification (SRS)
  - Software Design Document (SDD)
  - Software Project Management Plan (SPMP)
  - Software Test Document (STD)
- Installation of software program on target hardware
- Instructional materials need for the orientation of affected users

1.1.4. Schedule and Budget Summary

This is the schedule summary of the project. On the other hand, no budget was provided during the conduct of the study

Milestone Task	Date Started	Completion Date
Project Proposal	April 27, 2019	April 27, 2019
Data Gathering	June 3, 2019	June 28, 2019
Development Phase I	July 1, 2019	August 16, 2019
Software Test I	August 17, 2019	August 17, 2019
Development Phase II	August 19, 2019	September 27, 2019
Software Test II	September 28, 2019	September 28, 2019
Development Phase III	September 29, 2019	October 13, 2019

Table 1.0 Milestone Task

1.2. Evolution of plan

All changes to the project management plan must be agreed to by the project manager before they are implemented. All changes should be documented in order to keep the project management plan correct and up to date.

1.3. Definition, Acronyms, and Abbreviations

Term	Definition
TMC	Trinidad Municipal College
LGU	Local Government Unit
MVC Framework	**Model-View-Controller Framework**. Is a software architectural pattern commonly used for developing user interfaces which divides the related program logic into three interconnected elements.
LAN	**Local Area Network.** A computer network that links devices within a building or group of adjacent buildings.
RFID	**Radio Frequency Identification.** Is a form of wireless communication that incorporates the use of electromagnetic or electrostatic coupling in the radio frequency portion of the electromagnetic spectrum to uniquely identify an object, animal or person.

Table 2.0 Definition, Acronyms, and Abbreviations

2. References

[1] Analee E. Mayo, Public High Schools Online Library System PHOLS, February 2016

[2] S. M.Mojapelo and D. Luyanda, “Information access in high school libraries in limpopo province,south africa,” South African Journal of Libraries and Information Science, vol. 80, no. 2, 2014. http://sajlis.journals.ac.za/pub/article/view/1401/1422 Accessed 17 July 2015.

[3] ALA, “Types of libraries.” Web. http://www.ala.org/educationcareers/careers/librarycareerssite/typesoflibraries. Accessed 22 July 2015.

[4] By Hugh E. Williams, David Lane, Web Database Applications with PHP and MySQL: Building Effective Database, 2004

[5] RFID Technology, https://www.computer.org/csdl/magazine/pc/2006/01/b1025/13rRUxlgxLD

[6] IEEE Std 830-1998 (Revision of IEEE Std 830-1993)

[7] IEEE Std 1058-1998 (Revision and redesignation of IEEE Std 1058.1-1987, incorporating IEEE Std 1058-1998 and IEEE Std 1058a-1998)

3. Project Organization

This clause of the SPMP shall identify interfaces to organizational entities external to the project; describe the project's internal organizational structure; and define roles and responsibilities for the project.

3.1. External Structure

Library Automate is created for the Trinidad Municipal College. It will be utilized by the Librarian, Assistant Librarian, and its staff. The organizational chart of the external structure is presented below. Internal Structure

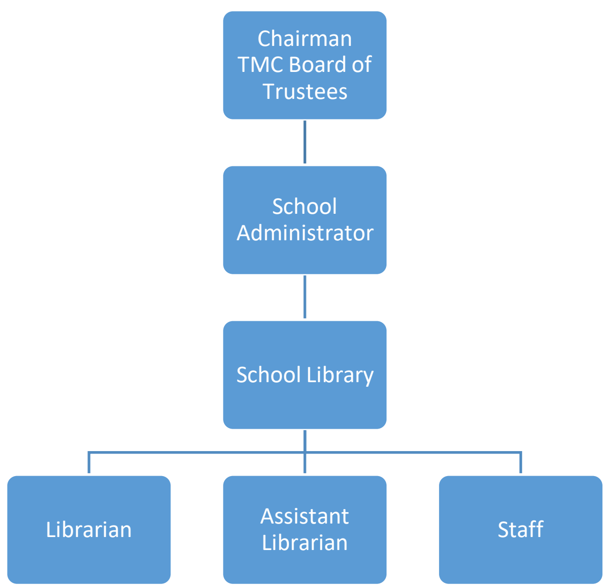

Figure 1.0 External Structure

3.2. Internal Structure

The Library Automate project is composed of one man development team that assumes the roles of Project Manager, Team Lead, Agile DBA, Developer, Software Tester, Front End Designer and Researcher. The lines of authority and communication of the aforementioned units are depicted in Figure 2.

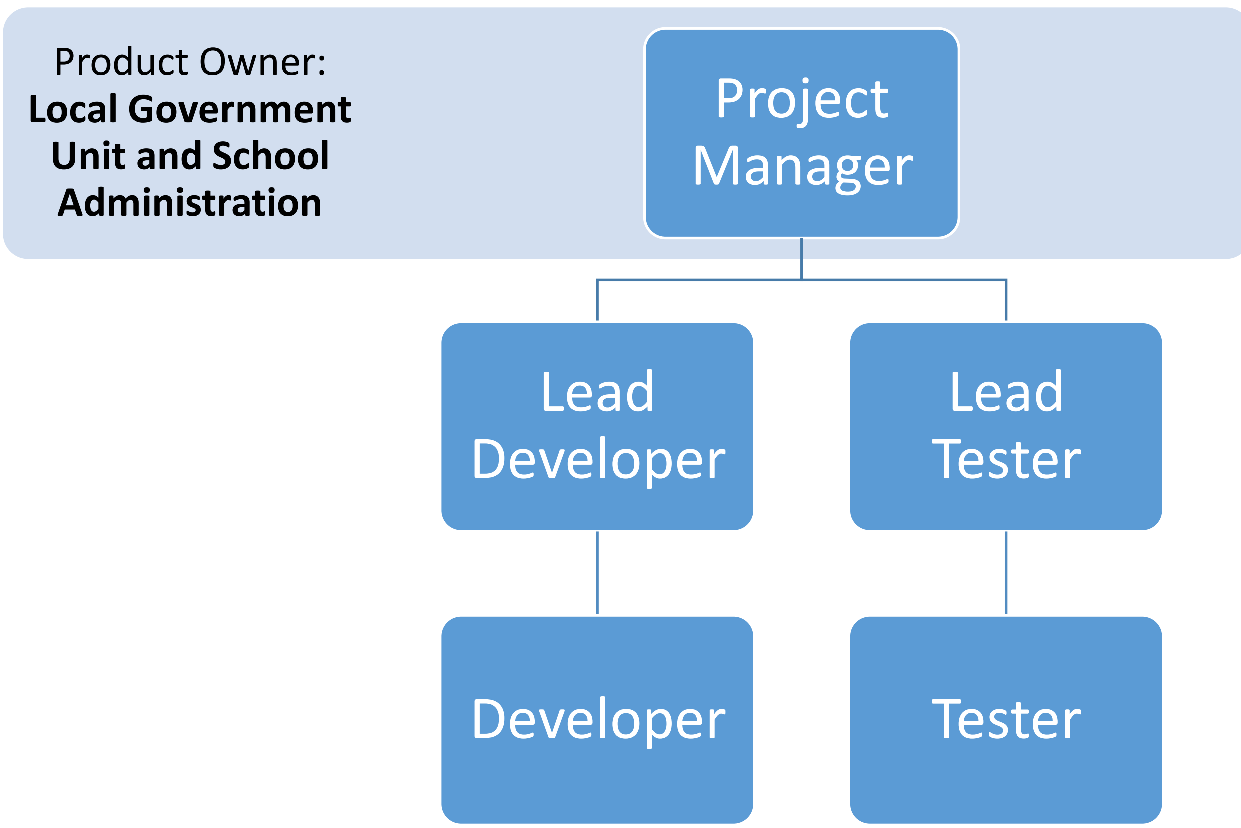

Figure 2.0 Internal Structure

3.3. Roles and Responsibilities

This section of the document describes the role and responsibilities for each member in an agile team that will be involve in project to deliver the Library Automate software requirement.

Role	Responsibilities
Product Owner	Monitor product backlog User acceptance test and Sign-off
Lead Developer	Source code management Code review and integration
Lead Tester	Test case management Test case script review
Developer	Requirement implementation Unit test
Tester	Test case script execution

Table 3.0 Roles and Responsibilities

4. Managerial Process Plans

This clause of the SPMP shall specify the project management processes for the project. This clause shall be consistent with the statement of project scope and shall include the project start-up plan, risk management plan, project work plan, project control plan, and project closeout plan.

4.1. Start-up plan

This clause of the SPMP shall specify the estimation, staffing, resources acquisition, and project staff training plan.

4.1.1. Estimation Plan

The proponent will be using an Agile Software Development since the nature of requirements defined by the user is dynamic. Constraints and rules are bound to change in the middle of the development and even after the deployment of the project. The development will be divided into three iterations but might expand if there are additional changes after these iterations.

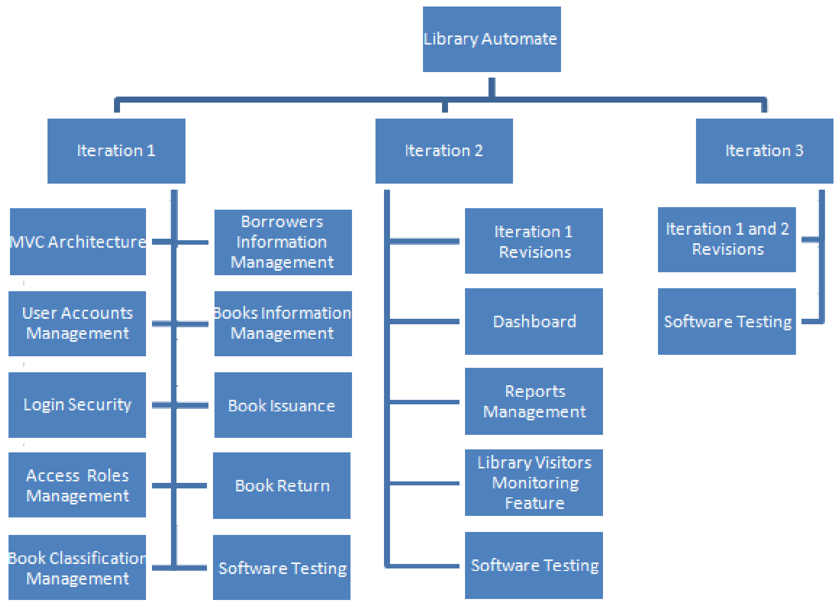

Figure 3.0 Feature Breakdown Structure

4.1.2. Staffing Plan

The table below shows the team member involved in Library Automate project.

Name	Position	Status
Local Government Unit of Trinidad & School Administration	Product Owner	Full Time
Clark Kevin V. Villamor	Project Manager	Full Time
Clark Kevin V. Villamor	Lead Developer	Full Time
Clark Kevin V. Villamor	Lead Tester	Full Time

Table 4.0 Staff and Person Involved in Project

4.1.3. Resource Acquisition Plan

All necessary hardware, software and tools for the project are already available. For requirements needed to gather, the project team will provide request letter to the librarian to access information from the library. The proponent will conduct informal interviews and list all information gathered.

4.1.4. Project Staff Training Plan

No training for Library Automate project participants will be needed. The project team composed of one-man team only but already well-trained and has years of experience in an agile way of working.

4.2. Work plan

This section describes in detail the decomposition of the total scope of work of the Library Automate project to be carried out by the project team to accomplish the project objectives and create the required deliverable.

Project Task		Estimated Schedule
1	**Iteration 1**	
1.1	MVC Architecture	July 1 – 5, 2019
1.2	User Accounts Management	July 6 – 9, 2019
1.3	Login Security	July 10, 2019
1.4	Access Roles Management	July 11-14, 2019
1.5	Book Classification Management	July 15-17, 2019
1.6	Borrowers Information Management	July 18-22, 2019
1.7	Books Information Management	July 23-28, 2019
1.8	Book Issuance	July 29 – August 4, 2019
1.9	Book Return	August 5-11, 2019
1.10	Software Testing	August 12-17, 2019
2	**Iteration 2**	
2.1	Iteration 1 Revisions	August 19 – September 6, 2019
2.2	Dashboard	September 7-10, 2019
2.3	Reports Management	September 11-17, 2019

2.4	Library Visitors Monitoring Feature	September 18-24, 2019
2.5	Software Testing	September 25-28, 2019
3	**Iteration 3**	
3.1	Iteration 1 and 2 Revisions	September 29 – October 6, 2019
3.2	Software Testing	October 7-11, 2019

Table 5.0 Work Plan

4.2.1. Work activities

The work activity table specifies the different chore per iteration tasks. It also shows the entire person assigned to the task and the number of hours allotted for each activity

Project Task		**Role**	**Labor Hours**
1	**Iteration 1**		
1.1	MVC Architecture		40
	Kick-off Meeting	Project Manager	2
	Requirements Gathering	Project Manager	8
	MVC Structure Development	Lead Developer	20
	Software Test	Lead Tester	10
1.2	User Accounts Management		18
	Kick-off Meeting	Project Manager	1
	Requirements Gathering	Project Manager	2
	Database Design	Lead Developer	3
	User Interface Design	Lead Developer	4
	CRUD Operation	Lead Developer	6
	Software Test	Lead Tester	2
1.3	Login Security		8
	Kick-off Meeting	Project Manager	1
	Database Design	Lead Developer	1
	User Interface Design	Lead Developer	2

	Login Development	Lead Developer	3
	Software Test	Lead Tester	1
1.4	Access Roles Management		18
	Kick-off Meeting	Project Manager	1
	Requirements Gathering	Project Manager	2
	Database Design	Lead Developer	3
	User Interface Design	Lead Developer	4
	CRUD Operation	Lead Developer	6
	Software Test	Lead Tester	2
1.5	Book Classification Management		18
	Kick-off Meeting	Project Manager	1
	Requirements Gathering	Project Manager	2
	Database Design	Lead Developer	3
	User Interface Design	Lead Developer	4
	CRUD Operation	Lead Developer	6
	Software Test	Lead Tester	2
1.6	Borrowers Information Management		40
	Kick-off Meeting	Project Manager	2
	Requirements Gathering	Project Manager	5
	Database Design	Lead Developer	8
	User Interface Design	Lead Developer	10
	CRUD Operation	Lead Developer	10
	Software Test	Lead Tester	5
1.7	Books Information Management		48
	Kick-off Meeting	Project Manager	2
	Requirements Gathering	Project Manager	8
	Database Design	Lead Developer	8
	User Interface Design	Lead Developer	10
	CRUD Operation	Lead Developer	15

	Software Test	Lead Tester	5
1.8	Book Issuance		56
	Kick-off Meeting	Project Manager	2
	Requirements Gathering	Project Manager	10
	Database Design	Lead Developer	10
	User Interface Design	Lead Developer	10
	CRUD Operation	Lead Developer	15
	Software Test	Lead Tester	9
1.9	Book Return		56
	Kick-off Meeting	Project Manager	2
	Requirements Gathering	Project Manager	10
	Database Design	Lead Developer	10
	User Interface Design	Lead Developer	10
	CRUD Operation	Lead Developer	15
	Software Test	Lead Tester	9
1.10	Software Testing		48
	Kick-off Meeting	Project Manager	2
	Software Test	Lead Tester	44
	Approval	Project Manager	2
2	**Iteration 2**		
2.1	Iteration 1 Revisions		152
	Kick-off Meeting	Project Manager	2
	Development Phase	Lead Developer	130
	Software Test	Lead Tester	20
2.2	Dashboard		32
	Kick-off Meeting	Project Manager	2
	Requirements Gathering	Project Manager	4
	User Interface Design	Lead Developer	12
	Retrieve Operation	Lead Developer	10

	Software Test	Lead Tester	4
2.3	Reports Management		56
	Kick-off Meeting	Project Manager	2
	Requirements Gathering	Project Manager	4
	User Interface Design	Lead Developer	20
	Retrieve Operation	Lead Developer	25
	Software Test	Lead Tester	5
2.4	Library Visitors Monitoring Feature		56
	Kick-off Meeting	Project Manager	2
	Requirements Gathering	Project Manager	4
	Database Design	Lead Developer	10
	User Interface Design	Lead Developer	5
	Hardware Equipment (RFID Reader/Writer) Integration	Lead Developer	25
	Software Test	Lead Tester	10
2.5	Software Testing		32
	Kick-off Meeting	Project Manager	2
	Software Test	Lead Tester	28
	Approval	Project Manager	2
3	**Iteration 3**		
3.1	Iteration 1 and 2 Revisions		64
	Kick-off Meeting	Project Manager	2
	Development Phase	Lead Developer	52
	Software Test	Lead Tester	10
3.2	Software Testing		40
	Kick-off Meeting	Project Manager	2
	Software Test	Lead Tester	36
	Approval	Project Manager	2

Table 6.0 Work Activities

4.2.2. Schedule Allocation

The project duration is constrained to three different iterations. Each iteration is allocated with different number of working hours due to different tasks for every iteration. However, schedule presented may change depending on the modifications of the requirements of the user

4.2.3. Resource Allocation

The development team has a fixed amount of time available for the project. The whole development period is divided into three iterations. Thus, the development team is given 350 hours for first iteration, 328 hours for second iteration, and 104 hours for third iteration to comply with the requirements and tasks assigned to them.

4.2.4. Budget Allocation

No budget was funded by stakeholder during the development phase. Thus, no necessary plan needed for allocating budget.

4.3. Control plan

This sub clause of the SPMP shall specify the metrics, reporting mechanisms, and control procedures necessary to measure, report, and control the product requirements, the project schedule, budget, and resources, and the quality of work processes and work products. All elements of the control plan should be consistent with the organization's standards, policies, and procedures for project control as well as with any contractual agreements for project control.

4.3.1. Requirements Control Plan

Requirement will be managed in use case description of SRS as requirements are changed.

4.3.2. Schedule Control Plan

The development team is going to have a daily stand up meeting to monitor their daily progress. The project manager would supervise the meeting. Every member of the team must identify their progress, impediments and the task they plan to accomplish the next day. In this way, the whole team knows the status of every developer. Moreover, they can learn on the different impediments mentioned by the other develop. Furthermore, the development team will conduct a retrospection meeting to evaluate whether the team has a good output for the current iteration. Practices that must be improved is then identified to enhance team's performance.

4.3.3. Budget Control Plan

As stated under section 4.2 sub section 4.2.4, no budget was provided for the development of the project.

4.3.4. Quality Control Plan

The tester will generate a separate Software Testing Document. From this document, checklists and other evaluation measures will be determined necessary or otherwise. Unit testing will be the main mechanisms of the development team to control the quality of the work process and the resulting work. More detailed plan is presented at Software Testing Document.

4.3.5. Reporting Plan

The team will submit a report to the end user every end of the iteration. This is to give update for the progress of development. During the development, the team will have a daily stand meeting which will serve as their report and updates for the accomplished tasks

4.3.6. Metrics Collection Plan

As mentioned in the schedule control plan, there will be daily stand up meeting to track the progress and impediments of every member. Also, a retrospection meeting is done every after iteration.

4.3.7. Risk Management Plan

Risks will be identified at the beginning of each phase and the team lead will assemble them into a prioritized risks list. During the daily stand up meeting, the team members will raise risks and reassess the prioritized risks. Team members will determine mitigation plans for all identified risks and tasks that need to be completed and then these risks and tasks will be assigned as action items. The team will monitor high priority risks every week. All risks will be documented by the team.

4.3.8 Project Closeout Plan

The development team will ensure the proper closeout of the project in the first semester of the academic year 2019-2020.

5. Technical Process Plans

This clause of the SPMP shall specify the development process model, the technical methods, tools, and techniques to be used to develop the various work products; plans for establishing and maintaining the project infrastructure; and the product acceptance plan.

5.1. Process Model

An agile project management methodology or framework used primarily for software development projects with the goal of delivering new software capability. The project methodology has iteration, once a week regular iteration review to present the work completed, and most important is the daily stand-up meeting for updates.

5.2. Methods, Tools, and Techniques

The methods and techniques listed in this table will be evaluated and applied in specific areas of the project as appropriate:

Category	Methods and Techniques
Requirements Elicitation	Meetings, Interviews, Brainstorming
Formal Specification and Analysis	Formal models using UML to model structural aspects of the requirements and design Use cases to define requirements
Protoype	Paper prototype

Table 7.0 Methods and Techniques

Category	Tools
Operating System	Windows 7 Ultimate
Programming Language	Hypertext Preprocessor C#.net
Database	MySQL
Front-end Design	Boostrap Framework
Diagrams	Astah Application
Hardware Equipment	RFID reader/writer and card
Documents	All documents will be written using Microsoft Word

Table 8.0 Tools

5.3. Infrastructure Plan

No infrastructure plan is defined in this project.

5.4. Product Acceptance Plan

Representative from the clients will be presented with Project Acceptance Form wherein the in-charge personnel of the proposed project will sign as a proof of accepting the final project.

6. Supporting Process Plans

This clause of the SPMP shall contain plans for the supporting processes that span the duration of the software project. These plans shall include, but are not limited to verification and validation, software documentation, quality assurance, and problem resolution. Plans for supporting processes shall be developed to a level of detail consistent with the other clauses and sub clauses of the SPMP. In particular, the roles, responsibilities, authorities, schedule, budgets, resource requirements, risk factors, and work products for each supporting process shall be specified. The nature and types of supporting processes required may vary from project to project; however, the absence of a configuration management plan, verification and validation plan, quality assurance plan, joint acquirer-supplier review plan, problem resolution plan, or subcontractor management plan shall be explicitly justified in any SPMP that does not include them. Plans for supporting processes may be incorporated directly into the SPMP or incorporated by reference to other plans.

6.1. Verification and Validation Plan

The development team has created a separate Software Testing Document for the verification and validation.

6.2. Documentation Plan

There are a number of documents that will be produced during the lifetime of the project. All documents are responsibility of the project team members. The lists of documents that will be created and maintained under version control include:

- Software Project Management Plan (SPMP)
- Software Requirements Specification (SRS)
- Software Testing Document (STD)
- Software Design Description (SDD)

6.3. Quality Assurance Plan

To ensure the quality of product, results from software test document shall be evaluated to distinguish all problems encountered. Problems found shall be fixed then.

6.4. Problem Resolution Plan

Any major problem faced by the development team should be discussed during the daily stand up meeting. Decisions are to be finalized by the project manager.